
KTU
NOTES

Soln:

$$\begin{bmatrix} 0 & 1 & 0 \\ -1 & 0 & -4 \\ 0 & 4 & 0 \end{bmatrix} \sim \begin{bmatrix} -1 & 0 & -4 \\ 0 & 1 & 0 \\ 0 & 4 & 0 \end{bmatrix} \quad R$$

$$\sim \begin{bmatrix} -1 & 0 & -4 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \quad R_3$$

$\therefore$ Rank $= 2$

2. Find the Eigen values of the matrix $A =$ [

what are the eigen values of A^2, A^{-1} w

it characteristic equation.

Soln:

trace of $A = \lambda_1 + \lambda_2$

$\therefore \lambda_1 + \lambda_2 = 4$ —— ①

$|A| = \lambda_1 \lambda_2 = 7 \quad \lambda_1 \lambda_2 = 3$ —

From eqn ① and ② $\lambda_1 = 1, \quad \lambda_2 = 3$

Eigen values of A are $1, 3$

Eigen values of A^2 are $1^2, 3^2$ (

Eigen values of A^{-1} are $1, 1/3$

$$\sim \begin{bmatrix} 1 & 2 & -1 & 3 \\ 0 & -7 & 5 & -8 \\ 0 & -6 & 5 & -4 \end{bmatrix}$$

$$\sim \begin{bmatrix} 1 & 2 & -1 & 3 \\ 0 & -7 & 5 & -8 \\ 0 & 0 & 5/7 & 20/7 \end{bmatrix} \quad R_3$$

rank A = rank $\tilde{A}$ = no. of unknowns $= 3$

$\therefore$ System of equation have unique solu

By Back substitution, $\frac{5}{7}z = \frac{20}{7}$

$-7y + 5z = -8 \Rightarrow 5z$

$x + 2y - z = 3 \Rightarrow x +$

$\Rightarrow x =$

$\therefore$ Solution is $x = -1, \ y = 4$

11 b) Find the Eigen values and Eigen vectors

$A - 0I = A \quad \begin{bmatrix} -6 \\ 2 \end{bmatrix}$

$$\sim \begin{bmatrix} 8 & -6 & 2 \\ 0 & 20/8 & \frac{-10}{4} \\ 0 & \frac{-10}{4} & \frac{5}{2} \end{bmatrix} \quad \begin{matrix} R_2 \\ R_3 \end{matrix}$$

$$\sim \begin{bmatrix} 8 & -6 & 2 \\ 0 & 20/8 & -10/4 \\ 0 & 0 & 0 \end{bmatrix} \quad R_3$$

By Back substitution $\frac{20}{8} x_2 -$

$x_2 = x_3$

Let $x_2 = 2 \Rightarrow x_3 = 2$

$8x_1 - 6x_2 + 2x_3 = 0$

$8x_1 = 6x_2 - 2x_3 = 6x_2 - 2x_3 \; 8$

Eigen vector corresponding to -1

$$X_1 = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}$$

$$\sim \begin{bmatrix} 5 & & \\ 0 & -16/5 & -8/5 \\ 0 & -8/5 & -4/5 \end{bmatrix}$$

$$\sim \begin{bmatrix} 5 & -6 & 2 \\ 0 & -16/5 & -8/5 \\ 0 & 0 & 0 \end{bmatrix} \quad R$$

By Back substitution $-\frac{16}{5} x$

$-2x_2$

Let $x_2 = 1$

$x_3 = -2$

$5x_1 - 6x_2 + 2x_3 = 0$

$5x_1 = 6x_2 - 2x_3 =$

$x_1 = 2$

Eigen vector corresponding to

$$X_2 = \begin{bmatrix} 2 \\ 1 \\ -2 \end{bmatrix}$$

$$\left[\begin{matrix} 0 & -\frac{40}{7} & -\frac{80}{7} \end{matrix}\right]$$

$$\sim \begin{bmatrix} -7 & -6 & 2 \\ 0 & -20/7 & -40/7 \\ 0 & 0 & 0 \end{bmatrix}$$

By Back substitution,

$$-\frac{20}{7}x_2 - \frac{40}{7}x_3$$

$$-x_2 - 2x_3 = 0 \Rightarrow$$

let $x_3 = 1$

$$x_2 = -2$$

$$-7x_1 - 6x_2 + 2x_3 = 0$$

$$7x_1 = -6x_2 + 2x_3 =$$

$$x_1 = 2$$

Eigen vector corresponding to $\lambda =$

$$X_3 = \begin{bmatrix} 2 \\ -2 \\ 1 \end{bmatrix}$$

$\therefore$ matrix of transformation that diagon

$$P = \begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & -2 \\ 2 & -2 & 1 \end{bmatrix}$$

Diagonal matrix $D = P^{-1}AP =$

$$\sim \begin{bmatrix} 2 & -2 & 4 & 0 & 0 \\ 0 & 0 & 0 & 5 & 15 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \quad \begin{matrix} R_2 \to R_2 - 2 \\ R_3 \to R_3 - \frac{1}{2} \end{matrix}$$

$\therefore$ Rank A = Rank $\tilde{A} = 2 <$ no. of va

$\therefore$ System of equations have infi

Solutions. By Back substitution,

$5w = 15 \Rightarrow w = 3$

$2x - 2y + 4z = 0$

let $y = t_1$, $z = t_2$

$2x = 2y - 4z = 2t_1 - 4t_2$

$x = t_1 - 2t_2$

$\therefore$ Solutions are $x = t_1 - 2t_2$, y

Eigen values are $\lambda =$

Eigen vector corresponding to

consider $(A-\lambda I)x = 0$

$$A + I = \begin{bmatrix} -4 & 2 \\ 2 & -1 \end{bmatrix}$$

$$\sim \begin{bmatrix} -4 & 2 \\ 0 & 0 \end{bmatrix}$$

By Back substitution, $-4x_1$

$4x_1 =$

Let $x_1 = 1$

Eigen vector corresponding to

$$X_1 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

By Back substitution,

$x_1 + 2x_2 = 0 \Rightarrow x_1 =$

let $x_2 = 1$ $x_1 =$

Eigen vector corresponding to $-1 = -$

$$X_2 = \begin{bmatrix} -2 \\ 1 \end{bmatrix}$$

The matrix of transformation th

matrix $P = \begin{bmatrix} 1 & -2 \\ 2 & 1 \end{bmatrix}$

$$D = P^{-1}AP = \begin{bmatrix} -1 & 0 \\ 0 & -6 \end{bmatrix}$$

$$z_x = \frac{y^3 - x^2 y}{(x^2+y^2)^2}.$$

$$\frac{\partial z}{\partial y} = \frac{x^3 - y^2 x}{(x^2+y^2)^2}.$$

$$= c^2 \sin(x-ct) \longrightarrow (1)$$

$$u_x = \cos(x-ct) \times 1$$

$$u_{xx} = -\sin(x-ct) \longrightarrow (2)$$

From eqn (1) and (2)

$$\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2}$$

13. a) The length and width of a rectan
measured with errors of at most
4% respectively. Use differentials
mate the max. % error in the cal

b) Find the local linear approxim
of $(x,y,z) = xyz$ at the point P(
Compute the error in approxima

$$A_x = y \quad , \quad A_y = x$$

$$A_x(x_0 y_0) = y_0$$

$$A_y(x_0 y_0) = x_0$$

$$dA = y_0\,dx + x_0\,dy$$

If $dx = \Delta x$ and $dy = \Delta y$

$$\Delta A \approx dA$$

$$\Delta x = x - x_0$$

$$\Delta y = y - y_0$$

$$\left|\frac{\Delta x}{x_0}\right| \leq 0.03$$

$$\left|\frac{\Delta y}{y_0}\right| \leq 0.04$$

$$\frac{\Delta A}{A_0} = \frac{y_0\,\Delta x + x_0\,\Delta y}{x_0\,y_0}$$

$\therefore$ The max. % error is area

$= 7\%$

b) $L(x,y,z) = f(x_0, y_0 z_0) + f_x(x_0 y_0 z_0)(x - x_0)$

$(y - y_0) + f_z(x_0 y_0 z_0)(z - z_0)$

$(x_0 y_0 z_0) = 1, 2, 3$

$f(1,2,3) = 6$

$f_x = yz = 6$

$f_y = xz$

$= 3$

$f_z = xy = 2$

$L(x,y,z) = 6 + 6(1.001 - 1) + 3(2.002 - 2)$

$= 6.018$

Error $= 0.018$

$$= \sqrt{(1.001-1)^2 + (2.002-2)^2 + (3.0\ldots}$$

$$= \sqrt{0.01}$$

$\therefore$ Error is less than distance b…

14 a) Let f be a differentiable funct

variables and suppose that

$w = f(x-y,\, y-z,\, z-x)$, show that

$$\frac{\partial w}{\partial x} = 0$$

b) Locate all relative extrema of

$$f(x,y) = 4xy - y^4 - x^4$$

$$\frac{\partial w}{\partial y} = \frac{\partial w}{\partial a} \times \frac{\partial a}{\partial y} + \frac{\partial w}{\partial b} \times \frac{\partial b}{\partial y}$$

$$= \frac{\partial w}{\partial a} \times -1 + \frac{\partial w}{\partial b} \times 1$$

$$= -\frac{\partial w}{\partial a} + \frac{\partial w}{\partial b} \longrightarrow (2)$$

$$\frac{\partial w}{\partial z} = \frac{\partial w}{\partial b} \times \frac{\partial b}{\partial z} + \frac{\partial w}{\partial c} \times \frac{\partial c}{\partial z}$$

$$= \frac{\partial w}{\partial b} \times -1 + \frac{\partial w}{\partial c} \times 1$$

$$= -\frac{\partial w}{\partial b} + \frac{\partial w}{\partial c} \longrightarrow (3)$$

From eqn (1), (2) and (3)

$$\frac{\partial w}{\partial x} + \frac{\partial w}{\partial y} + \frac{\partial w}{\partial z} = 0$$

$$f_x = 0$$

$$\Rightarrow 4y - 4x^3 = 0$$

$$y = x^3 \longrightarrow (1)$$

$$f_y = 0$$

$$\Rightarrow 4x - 4y^3 = 0$$

$$x = y^3 \longrightarrow (2)$$

Substitute eqn (1) in (2)

$$x = (x^3)^3$$

$$x = x^9$$

$$x - x^9 = 0$$

$$x(1 - x^8) = 0$$

$$x = 0$$

$$1 - x^8 = 0$$

$$x^8 = 1 \Rightarrow x = 1 \text{ or } -1$$

$$D = f_{xx}(x_0 y_0) f_{yy}(x_0 y_0) - f_{xy}^2 ($$

$$f_{xx} = -12x^2$$

$$f_{yy} = -12y^2$$

$$f_{xy} = 4$$

at $(0,0)$

$$D = (-12 \times 0)(-12 \times 0) - 4^2$$

$$= -16$$

$$D < 0$$

$\therefore$ $(0,0)$ is a saddle point.

$D > 0$

$\therefore f_x = -12$

$f_x < 0$

$\therefore$ f has relative maximum at (

at $(-1, -1)$

$$D = -12 \times (-1)^2 \times -12 \times (-1)^2 - 4^2$$

$$= 144 - 16$$

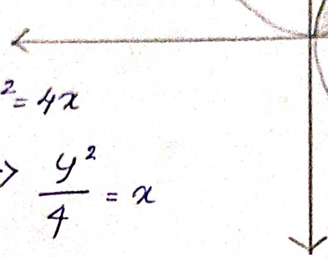
$$= 128$$

$D > 0$

$f_x = -12$, $f_x < 0$

$\therefore$ f has relative maximum at (

$y = 2x^2 , \; y = 2\sqrt{x}$



Given $x^2 = \frac{y}{2}$, and $y^2 = 4x$

$$\Rightarrow \frac{y^2}{4} = x$$

$$\frac{y^4}{16} = \frac{y}{2}$$

$$\Rightarrow y^3 = 8 \Rightarrow y = 2$$

$$x^2 = \frac{y}{2} \Rightarrow x^2 = 1 \Rightarrow x = 1$$

$\therefore$ limits of x are $x = 0$ and $x = 1$.

$$\text{Area} = \iint_R dA = \int_0^1 \int_{2x^2}^{2\sqrt{x}} dy\, dx = \int_0^1 [y]_{2x^2}^{2\sqrt{x}}$$

$$= \int_0^1 (2\sqrt{x} - 2x^2)\, dx$$

$$= 2\left[\frac{2}{3}\left[x^{3/2}\right]_0^1 - \frac{1}{3}\left[x^3\right]_0^1\right]$$

$$= 2\left[\frac{2}{3} - \frac{1}{3}\right] = \underline{\underline{\frac{2}{3}}}$$

$$= \frac{-1}{2}\int_0^{2\pi} \left[e^{-r^2}\right]_0^2 d\theta \quad = \frac{-1}{2}\int_0^{2\pi} \left[e^{-4} - e^0\right.$$

$$= -\frac{1}{2}\int_0^{2\pi} (e^{-4}-1)\, d\theta$$

$$= -\frac{1}{2}(e^{-4}-1)\,[\theta]_0^{2\pi}$$

$$= \pi(1-e^{-4})$$

16. a) Evaluate $\int_0^1\int_y^1 \frac{x}{x^2+y^2}\,dx\,dy$ by reversing the integration

b) Use triple integral to find the volume within the cylinder $x^2+y^2=9$ and planes $z=1$ and $x+z=5$.

Ans:- a) $\int_0^1 \int_y^1 \frac{x}{x^2+y^2}\, dx\, dy$

$$= \int_0^1 \int_0^x \frac{x}{x^2+y^2}\, dy\, dx$$

$$= \int_0^1 x \left[\frac{1}{x} \tan^{-1} \frac{y}{x}\right]_0^x dx$$

$$= \int_0^1 \frac{x}{x} \left[\tan^{-1} 1 - \tan^{-1} 0\right] dx$$

$$= \int_0^1 \frac{\pi}{4}\, dx = \frac{\pi}{4} [x]_0^1 = \frac{\pi}{4}$$

$$\int_0 \int_0^3 \int_1^{5-x} dz\, r\,dr\,d\theta$$

$$\Rightarrow \int_0^{2\pi}\int_0^3 [z]_1^{5-x}\, r\,dr\,d\theta \Rightarrow \int_0^{2\pi}\int_0^3 5-x-1\; r\,d$$

$$\Rightarrow \int_0^{2\pi}\int_0^3 (5-r\cos\theta-1)\, r\,dr\,d\theta \Rightarrow \int_0^{2\pi}\int_0^3 (5r-r^2\cos\theta -$$

$$\Rightarrow \int_0^{2\pi} \left[5\frac{r^2}{2}\right]_0^3 - \left[\cos\theta\frac{r^3}{3}\right]_0^3 - \left[\frac{r^2}{2}\right]_0^3 d\theta \Rightarrow \int_0^{2\pi}\left(\frac{45}{2}\right.$$

$$\Rightarrow \int_0^{2\pi} 18-9\cos\theta\; d\theta \Rightarrow \int_0^{2\pi} 18\,d\theta - \int_0^{2\pi} 9\cos\theta\, d$$

$$\Rightarrow \left[18[\theta]_0^{2\pi} - 9[\sin\theta]_0^{2\pi}\right] \Rightarrow \underline{36\pi}.$$

$= \dfrac{9}{2}$

6. Find the mass of the lamina with ... is bounded by the x-axis, $x = 1$, ...

ans:- $\displaystyle\int_0^1 \int_0^{\sqrt{x}} (x + 2y)\, dy\, dx$

$= \displaystyle\int_0^1 \left[x[y]_0^{\sqrt{x}} + [y^2]_0^{\sqrt{x}} \right] dx$

$= \displaystyle\int_0^1 \left(x^{3/2} + x \right) dx$

$= \dfrac{2}{5} \left[x^{5/2} \right]_0^1 + \left[\dfrac{x^2}{2} \right]_0^1 \quad = \dfrac{2}{5} + \dfrac{1}{2}$

$= \dfrac{9}{10}$

$\therefore 5.373737\ldots = 5 + \frac{37}{99} = \frac{532}{99}$

8. Examine the convergence of $\sum_{k}$

Ans: Consider $\sum \frac{k^2}{k^2} = \sum 1$ which is

$$\underset{k\to\infty}{Lt} \frac{a_k}{b_k} = \underset{k\to\infty}{Lt} \frac{k^2}{2k^2+3} = \underset{k\to\infty}{Lt}$$

$= 1 > 0$ & finite.

by limit comparison test, since

$\sum_{k=1}^{\infty} \frac{k^2}{2k^2+3}$ is also divergent.

$$b_k = \frac{3}{k^4}$$

$$\underset{k\to\infty}{Lt} \frac{a_k}{b_k} = \underset{k\to\infty}{Lt} \frac{3k^3 - 2k^2 + 4}{k^7 - k^3 + 2} \cdot \frac{k^4}{3}$$

$$= \underset{k\to\infty}{Lt} \frac{3k^7 - 2k^6 + 4k^4}{3(k^7 - k^3 + 2)}$$

$$= \underset{k\to\infty}{Lt} \frac{3k^7\left(1 - \frac{2k^6}{3k^7} + \frac{4}{3}\right.}{3k^7\left(1 - \frac{k^3}{k^7} + \frac{2}{k}\right.}$$

By limit comparison test Σa_k con

Since Σb_k is a convergent series by

ii) $a_k = \frac{k^k}{k!}$

$$a_{k+1} = \frac{(k+1)^{k+1}}{(k+1)!}$$

$$\underset{k\to\infty}{Lt} \frac{a_{k+1}}{a_k} = \underset{k\to\infty}{Lt} \frac{(k+1)^{k+1}}{(k+1)!} \frac{k!}{k^k} = \underset{k\to\infty}{Lt} \frac{(k}{}$$

$$= \underset{k\to\infty}{Lt} \left(\frac{k+1}{k}\right)^k = \underset{k\to\infty}{Lt} \left(1 + \frac{1}{k}\right)^k =$$

By ratio test the series diverges

$\therefore$ by Leibnitz test, the series conv

$$u_k = (-1)^{k+1} \frac{1}{\sqrt{k+1}}$$

$$|u_k| = \frac{1}{\sqrt{k+1}}$$

$$\therefore \quad v_k = \frac{1}{\sqrt{k}}$$

$$|u_k| < v_k$$

by limit comparison test, $\Sigma |u_k|$ div

$\Rightarrow \Sigma u_k$ diverges absolut

$\Rightarrow$ conditionally convergent.

18.a) Test the convergence of the serie

$$\frac{1}{1\cdot 2} + \frac{1}{2\cdot 3} + \frac{1}{3\cdot 4} + \frac{1}{4\cdot 5} + \cdots$$

Ans:

$$\frac{1}{1\cdot 2} + \frac{1}{2\cdot 3} + \cdots = \sum_{k=1}^{\infty} \frac{1}{k(k+1)}$$

Let $a_k = \frac{1}{k(k+1)}$, $b_k = \frac{1}{k^2}$

by root test $\underset{k\to\infty}{Lt}(a_k)^{1/k} = \underset{k\to\infty}{Lt}\left[\left(\frac{k}{k+\right.\right.$

$$= \underset{k\to\infty}{Lt}\left(\frac{k}{k+1}\right)^{k} =$$

$$= \underset{k\to\infty}{Lt}\frac{k^k}{k^k\left(1+\frac{1}{k}\right)^k}$$

$\therefore$ the series converges.

ii) $a_k = \dfrac{7^k}{k!}$

$a_{k+1} = \dfrac{7^{k+1}}{(k+1)!}$

by ratio test, $\underset{k\to\infty}{Lt}\dfrac{a_{k+1}}{a_k} = \underset{k\to\infty}{Lt}$

$$= \underset{k\to\infty}{Lt}\frac{7^{k+1}}{(k+1)!}\cdot\frac{k!}{7^k}$$

$$= \underset{k\to\infty}{Lt}\frac{7}{k+1} = 0 < 1$$

$\therefore \Sigma a_k$ converges.

$f(x) = \sin \pi x$

$f(1/2) = \sin \pi/2 = 1$

$f'(x) = \cos \pi x \times \pi$

$f'(1/2) = \pi \times \cos \pi/2$

$= 0$

$f''(x) = \pi \times \bar{\ } \sin \pi x \times \pi$

$= -\pi^2 \sin \pi x$

$f''(1/2) = -\pi^2 \sin \pi/2$

$= -\pi^2$

$f'''(x) = -\pi^3 \cos \pi x$

$f'''(1/2) = -\pi^3 \times \cos \pi/2$

$= 0$

$f^{IV}(x) = +\pi^4 \sin \pi x$

$f^{IV}(1/2) = \pi^4 \sin \pi/2$

$= \pi^4$

$\therefore f(x) = 1 + \frac{x - 1/2}{1!} \times 0 + \frac{(x-1/2)^2}{2!} \times -\pi^2 + \frac{(x-}{}$

$(x-1/2)^4/4!\ \pi^4$

$$\therefore \quad a_0 = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x)\, dx$$

$$a_n = \frac{1}{L} \int_{\alpha}^{\alpha+2L} f(x) \cos\left(\frac{n\pi x}{L}\right) dx$$

$$\therefore \quad a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos(nx)\, dx$$

$$b_n = \frac{1}{L} \int_{\alpha}^{\alpha+2L} f(x) \sin\left(\frac{n\pi x}{L}\right)$$

$$\therefore \quad b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin(nx)\, dx$$

$$a_0 = \frac{1}{2l}\int_{\alpha}^{\alpha+2l} f(x)\,dx$$

$$= \frac{1}{2}\int_{-1}^{1} x-x^2\,dx$$

$$= \frac{1}{2}\left[\frac{x^2}{2}-\frac{x^3}{3}\right]_{-1}^{1}$$

$$= \frac{1}{2}\left[\frac{1}{2}-\frac{1}{3}-\left(\frac{1}{2}-\frac{-1}{3}\right)\right]$$

$$= \frac{1}{2}\times\frac{-2}{3} = \frac{-1}{3}$$

$$a_n = \frac{1}{l}\int_{\alpha}^{\alpha+2l} f(x)\cos\left(\frac{n\pi x}{l}\right)dx$$

$$= \int_{-1}^{1}(x-x^2)\cos(n\pi x)\,dx$$

$$= (x-x^2)\frac{\sin(n\pi x)}{n\pi} - \Big[(1-2x)$$

$$+(-2).\left(\frac{-\sin(n\pi x)}{n^3\pi^3}\right)\Big]_{-1}^{1}$$

$$b_n = \frac{1}{l}\int_{-l}^{l} f(x)\ \sin\left(\frac{n\pi x}{l}\right) dx$$

$$= \int_{-1}^{1} (x-x^2)\ \sin(n\pi x)\ dx$$

$$= (x-x^2) . \frac{-\cos n\pi x}{n\pi} - (1-2x)$$

$$+ (-2) . \frac{\cos(n\pi x)}{n^3\pi^3} \Bigg]_{-1}^{1}$$

$$= \frac{-1}{n\pi}\left[0 - {}^{-}2 \times \cos n\pi\right] - 0 - \frac{2}{n^{3}}$$

$$= \frac{-2}{n\pi} \cos n\pi$$

$$= \frac{-2}{n\pi}(-1)^n$$

$$\therefore f(x) = \frac{-1}{3} + \sum_{n=1}^{\infty} \frac{-4}{n^2\pi^2}(-1)^n \cos\left(\frac{n\pi x}{l}\right)$$

$$= \frac{-1}{2}\left[e^{-2}-1\right]$$

$$a_n = \frac{2}{l}\int_0^l f(x)\cos\left(\frac{n\pi x}{l}\right)dx$$

$$= \frac{2}{2}\int_0^2 e^{-x}\cdot\cos\left(\frac{n\pi x}{2}\right)dx$$

$$= \frac{e^{-x}}{(-1)^2+\frac{n^2\pi^2}{4}}\left[\cancel{\frac{n\pi}{2}} - \cos\left(\frac{n\pi x}{2}\right) + \right.$$

$$= \frac{e^{-2}}{1+\frac{n^2\pi^2}{4}}\left[-\cos(n\pi) + \frac{n\pi}{2}\times \sin(n\right.$$

$$= \frac{e^{-2}}{1+\frac{n^2\pi^2}{4}}\left[(-1)^n\right] + \left(\frac{1}{1+\frac{n^2\pi^2}{4}}\right]$$

$$= \frac{1-e^{-2}(-1)^n}{1+\frac{n^2\pi^2}{4}}$$

$$\therefore f(x) = \frac{-1}{2}\left[e^{-2}-1\right] + \sum_{n=1}^{\infty}\frac{1-e^{-2}(-1)^n}{1+\frac{n^2\pi^2}{4}}$$

$$= \frac{2}{2l} \int_0^l f(x)\,dx \quad (\because f(x) \text{ i}$$

$$= \frac{1}{l} \int_0^l f(x)\,dx$$

$$= \frac{1}{\pi} \int_0^{\pi} x^2\,dx = \frac{1}{\pi} \left[\frac{x^3}{3}\right]_0^{\pi}$$

$$= \frac{\pi^2}{3}$$

$$a_n = \frac{1}{l} \int_{-l}^{l} f(x) \cos\left(\frac{n\pi x}{l}\right) dx$$

$$= \frac{2}{l} \int_0^l x^2 \cos\left(\frac{n\pi x}{\pi}\right) dx$$

$$= \frac{2}{\pi} \int_0^{\pi} x^2 \cos(nx)\,dx$$

$$= \frac{2}{\pi} \left[x^2 \times \frac{\sin(nx)}{n} - 2x \times \right.$$

$$2 \times \frac{-\sin n}{n^3}$$

$$= \frac{4}{n^2}(-1)$$

$$\therefore f(x) = \frac{\pi^2}{3} + \sum_{n=1}^{\infty} \frac{4}{n^2}(-1)^n \cos$$

$$x^2 = \frac{\pi^2}{3} + 4\left[-\frac{\cos x}{1^2} + \frac{\cos}{2}\right. \rightarrow (1)$$

Put $x = \pi$, in (1)

$$\pi^2 = \frac{\pi^2}{3} + 4\left[\frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2}\right.$$

$$\frac{2\pi^2}{3} = 4\left[\frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \cdots\right.$$

$$\frac{\pi^2}{6} = 1 + \frac{1}{2^2} + \frac{1}{3^2} + \cdots$$

When $x = 0$

$$0 = \frac{\pi^2}{3} + 4\left[-1 + \frac{1}{2^2} - \frac{1}{3^2} + \cdots\right]$$

$$= \frac{\pi^2}{3} - 4\left[1 - \frac{1}{2^2} + \frac{1}{3^2} - \cdots\right]$$

$$\frac{\pi^2}{12} = 1 - \frac{1}{2^2} + \frac{1}{3^2} - \cdots$$

20 b) Obtain the half range Fourie

of $f(x) = \begin{cases} x, & 0 < x < \pi/2 \\ \pi - x, & \pi/2 < x < \pi \end{cases}$

Soln: Fourier sine series for $f(x)$ i

$$f(x) = \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi x}{l}\right)$$

$$= \frac{2}{\pi}\left\{\left[\frac{-\frac{\pi}{2}\cos\left(\frac{n\pi}{2}\right)}{n} + \frac{1}{n^2}\sin\left(\frac{n\pi}{2}\right)\right.\right.$$

$$= \frac{2}{\pi}\,\frac{2}{n^2}\sin\frac{n\pi}{2}$$

$$= \frac{4}{\pi n^2}\sin\frac{n\pi}{2}$$

$$\therefore f(x) = \sum_{n=1}^{\infty}\frac{4}{n^2\pi}\sin\frac{n\pi}{2}\,\sin nx$$